Project Title: Exploratory data analysis

Objective: to gain a deeper understanding of the data's structure, identify patterns, and uncover insights that can inform subsequent data analysis and modeling tasks, ultimately aimed at predicting passenger survival.

Tools & sources:

Tools: python libraries such as pandas, matplotlib,seaborn for EDA.

Sources: Used titanic dataset.csv file from kaggle

Methodology/ steps:

1.Import python libraries such as:

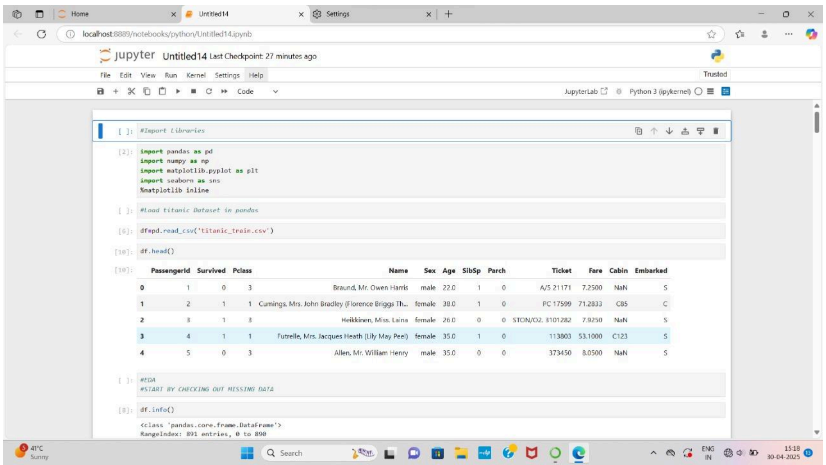
•pandas:used for reading csv file ,data cleaning, manipulation analysis.

•.seaborn:seaborn is a library for creating statistical graphics in Python.like charts,plot graphs.

•Matplotlib:used for creating interactive and animated visualization.

•%matplotlib inline:allows us to interact with your plots directly in the notebook or shell.

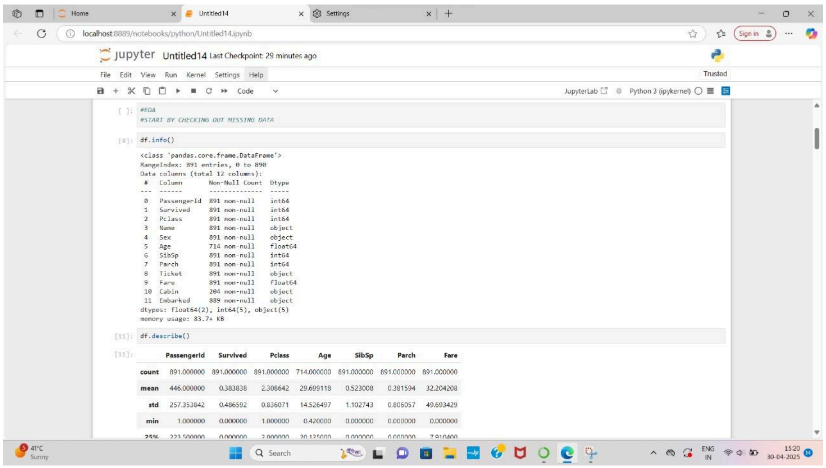
2.Load the dataset in pandas libraries to Read CSV file, as pd.read_csv().



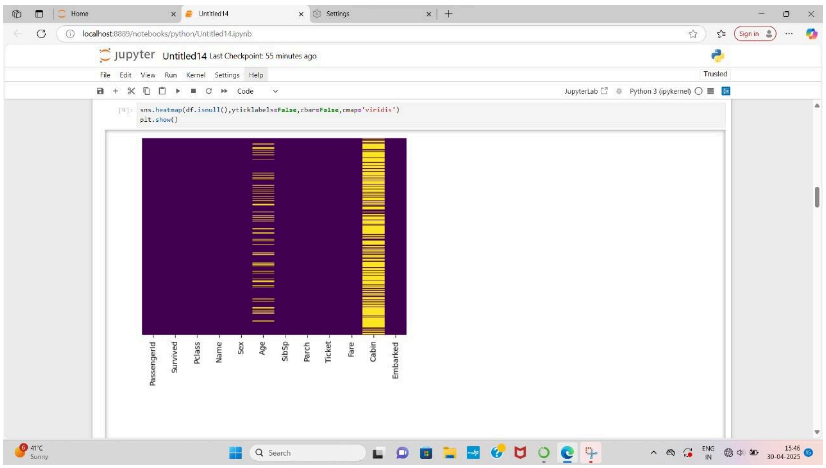
3.Exploratory data analysis:

Start by checking missing data such as

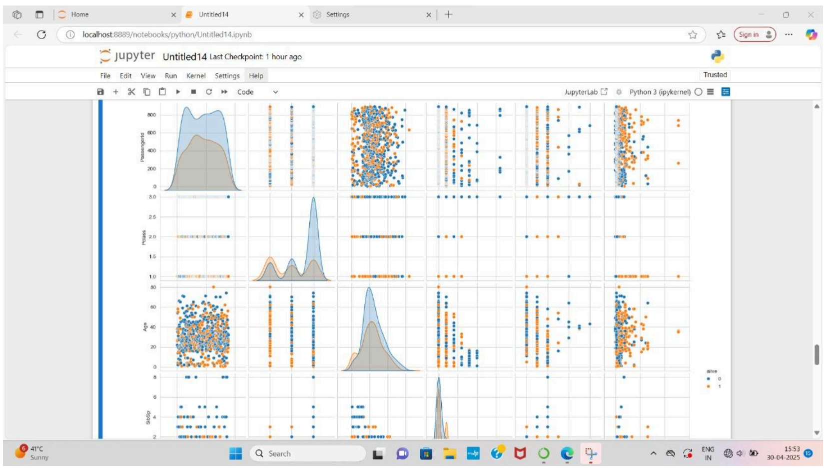
df.info(), df.isnull(), df.describe(),
df.count_values().



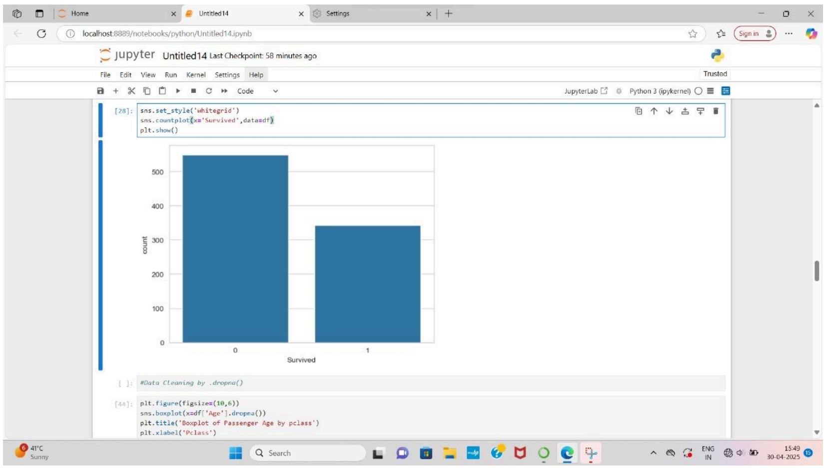
4. Used seaborn to create simple heatmap to see where are missing data



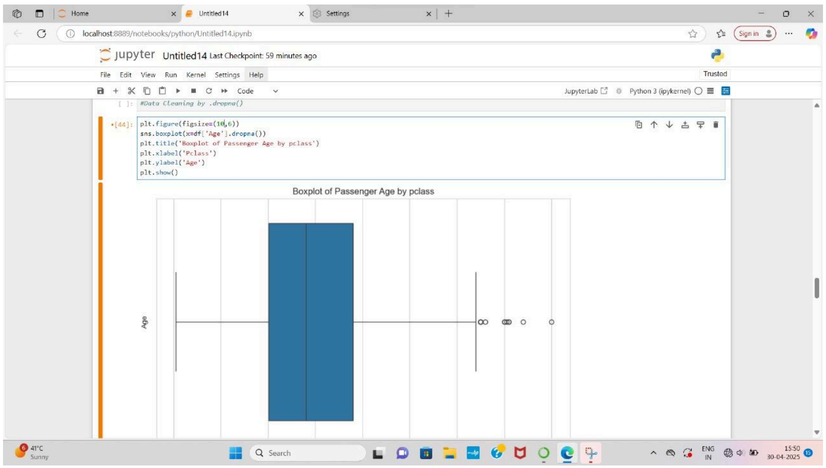
5.created a pairplot by alive columns to colour points by survival



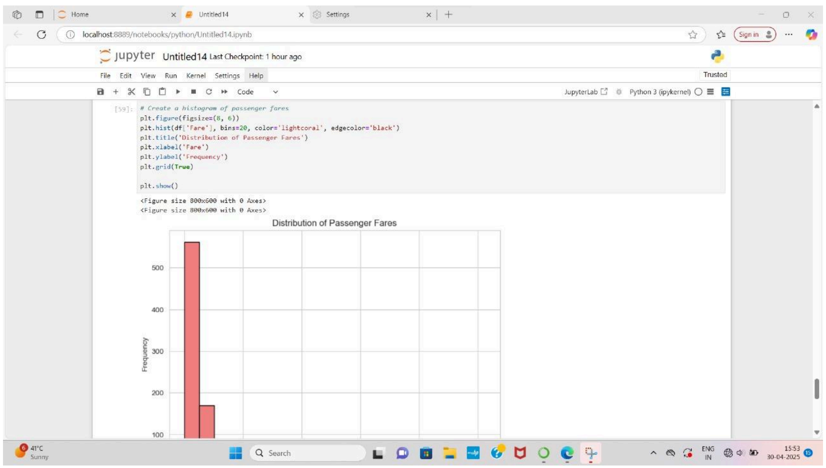
6. Created a countplot based on survived column where 0 indicate not survive while 1 indicate person is survive

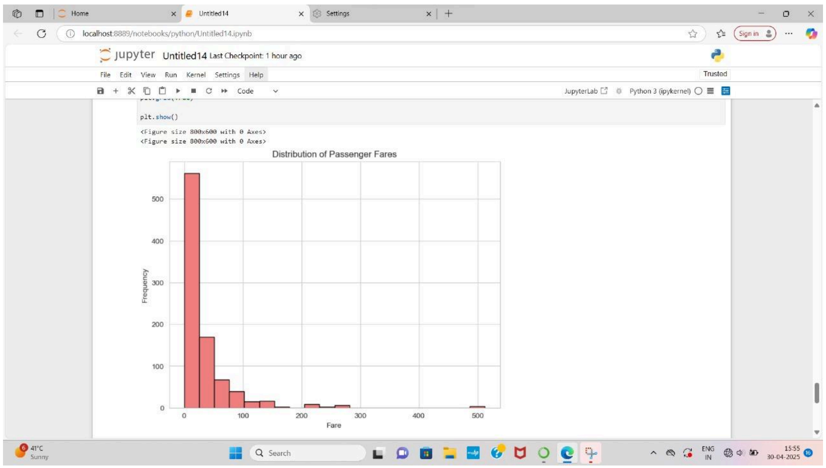


7.created boxplot of passenger age by pclass and data cleaning by drop.na()

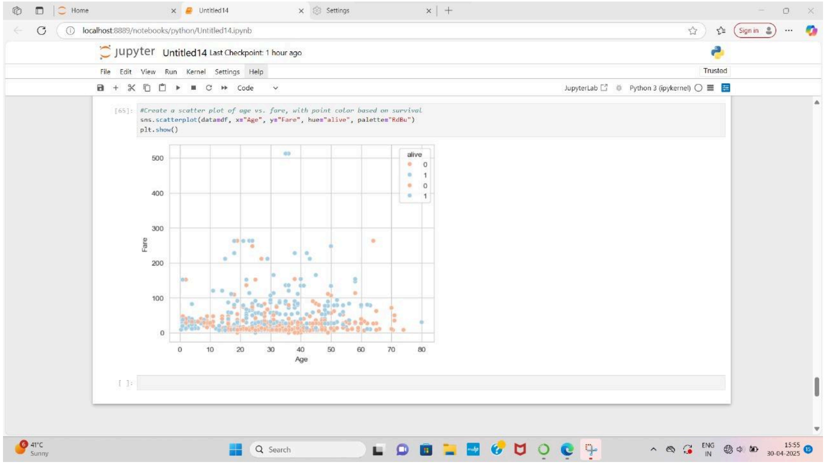


8.created histogram of passenger fares





9.created a scatterplot of age vs fare with point color based on survival



Conclusion & insights:

Analysis of the Titanic dataset reveals key factors influencing passenger survival.

Women had a significantly higher survival rate than men.

Passengers in higher classes (1st and 2nd) were more likely to survive than those in 3rd class.

Children also had a higher survival rate, while elderly passengers had lower chances.

The number of siblings/spouses and parents/children aboard did not show a strong correlation with survival.

This suggests that gender, passenger class, and age were the primary determinants of survival on the Titanic.